Phase 1, Module 3: Data Types & Data Integrity

Khyber AI Systems Engineer (KASE)

Welcome back. Today we look at the raw materials of our AI systems: Numbers and Text. But we are going to look at them like engineers, focusing on precision, memory, and formatting for production.

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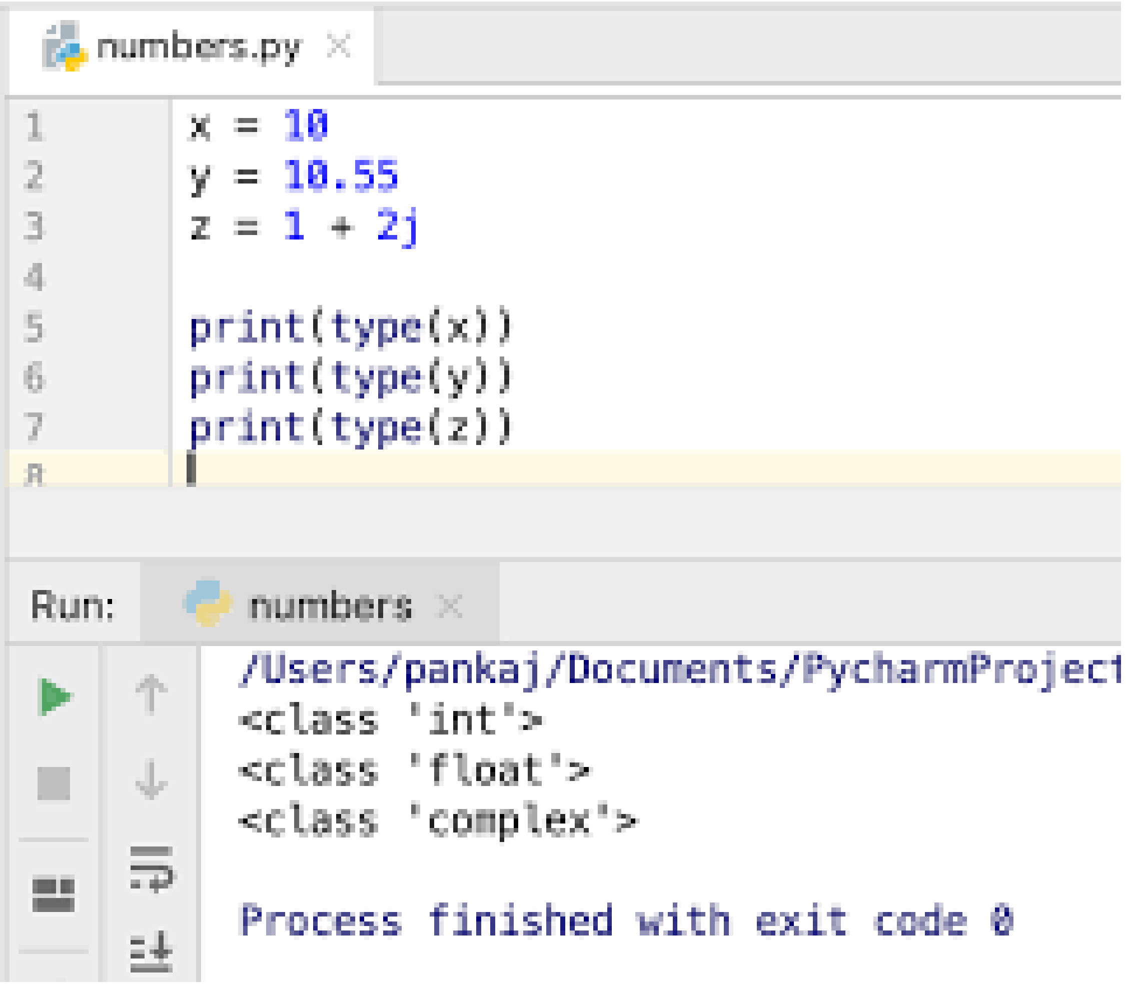
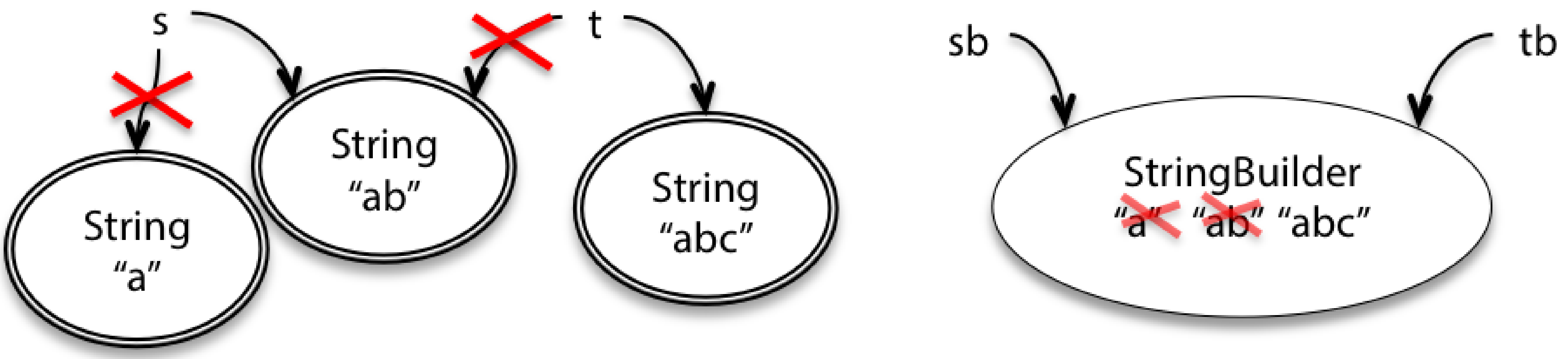
The Numerics (Integers vs. Floats)

The Danger of Decimals

- **Integers (int):** Whole numbers. In Python, they can be infinitely large.
- **Floats (float):** Decimal numbers. They are stored using the IEEE 754 standard.
- **The Trap:** Computers cannot store fractions perfectly in binary. 0.1 + 0.2 does NOT equal 0.3.





Strings as Immutable Sequences

Text is Carved in Stone

- A String (str) is just a sequence of characters in memory.
- Immutability: Once a string is created, you cannot change a single letter inside it.
- To "change" a string, Python actually destroys the old one and creates a completely new one in RAM.

Modern String Formatting

```
>>> greeting = "Hello World"
>>> name = "Zenon"
>>> planet = "X"
>>> intro = (
...     f"{greeting}. "
...     f"I am {name}. "
...     f"I come from {planet}."
... )
>>> print(intro)
"Hello World. I am Zenon. I come from X."
```

The 2026 Standard (F-Strings)

- Old Way (Pre-2015): %s or .format() (Slow and hard to read).
- Modern Way: F-Strings f"Hello {name}".
- Why? F-Strings are compiled directly in C, making them mathematically faster and much cleaner for API development.